MULTIMEDIA UNIVERSITY OF KENYA

Faculty of Computing and Information Technology

Department of Computer Science

CSE 2222: Data Communications and Computer Networks — Continuous Assessment Assignment

Course/Year:	BSC Software Engineering — Year 2, Semester 2 (Y2S2)	**Date:**	June 5, 2026
Task:	Individual Solutions Document	**Total Weight:**	30% of Total Course Marks (70 Marks Total)

Question 1

(10 Marks)

> *You are the administrator of a LAN that has increased the number of users by a factor of three over the past couple of years. You have been adding users by increasing the number of switches and switch ports on the network. Now users are complaining that network response is slowing and even their computers seem slower. You monitor the network for a few days and find that broadcast frames constitute a high percentage of your network traffic.*

a. What can you do to contain the broadcasts?

(4 Marks)

To effectively contain and eliminate the high volume of broadcast frames propagating across the single large flat network, the following strategic actions should be implemented:

- **Implement Virtual Local Area Networks (VLANs):** Break up the single, flat Layer 2 broadcast domain into multiple separate, logical broadcast domains (VLANs) based on departments, functions, or physical floors. Layer 2 switches do not forward broadcast frames between different VLANs.
- **Introduce a Layer 3 Routing Device:** Deploy a router or a Layer 3 multilayer switch to interconnect the newly formed VLANs. Layer 3 devices drop Layer 2 broadcast frames by default, preventing them from bleeding across subnets.
- **Configure Storm Control / Broadcast Suppression:** Enable storm control limits on switch access ports to drop broadcast traffic if it exceeds a specified threshold percentage of the port's total bandwidth, mitigating severe performance drops.
- **Optimize Protocols and Timers:** Disable unused dynamic broadcast-reliant protocols on hosts and switches (e.g., turning off legacy NetBIOS, optimizing ARP cache timeouts, or tuning DHCP lease timers where applicable).

b. What network reconfiguration might you have to do? Justify your decision. *(6 Marks)*

Proposed Reconfiguration: Transition from a flat flat Layer 2 topology into a *Segmented Hierarchical Layer 3 Network* using VLANs and Inter-VLAN Routing via a Multilayer Switch or Router-on-a-Stick architecture.

Justification Metrics:

1. **Segmentation of Broadcast Domains:** Standard Layer 2 switches flood broadcast frames (like ARP requests or DHCP discoveries) out of every single active port except the ingress port. As the host count tripled, the cumulative background broadcast traffic scaled exponentially. Each network interface card (NIC) must interrupt its host CPU to process every broadcast frame. This directly explains why users' physical computers feel slower. Segmenting into smaller subnets restricts broadcasts strictly to their originating local VLAN.
2. **Traffic Optimization and Bandwidth Reclamation:** By ensuring broadcasts are isolated, valuable link bandwidth is reclaimed across switch trunks, bringing down latency and accelerating network response times.
3. **Enhanced Security and Manageability:** Dividing users into functional subnets (e.g., Finance, Engineering, Guest) allows for the application of Access Control Lists (ACLs) at the Layer 3 boundary, protecting critical internal company data assets from unauthorized lateral visibility.

Question 2 *(10 Marks)*

> *You must install 125 computers for a new business that wants to run TCP/IP and have access to the Internet. The ISP in town will assign you only four public IP addresses, so you decide to assign the computers addresses in the range 172.16.1.1/16 through 172.16.1.125/16. What else must you do to allow these computers to access the Internet? Explain your reasons.*

To successfully connect these 125 computers to the Internet using only 4 available public IP addresses, you must implement and configure the following technical services and mechanisms:

1. **Configure Network Address Translation (NAT) / Port Address Translation (PAT):**
   - *Action:* Set up dynamic NAT with overload (commonly called PAT) on the border gateway router. Map the internal private IP pool (**172.16.1.1** to **172.16.1.125**) to the 4 public IP addresses provided by the ISP.
   - *Reasoning:* The **172.16.0.0/12** range consists of RFC 1918 Private IP addresses. Public Internet routers reject and drop private addresses automatically. PAT tracks outbound flows using distinct source TCP/UDP port numbers assigned to each private host, allowing thousands of concurrent internal sessions to safely share a handful of public IPs.

2. **Establish a Default Gateway Configuration:**
   - *Action:* Allocate a dedicated IP address on the internal interface of your core router within the local subnet (e.g., **172.16.0.1**) and configure it as the Default Gateway on all 125 client machines.

- *Reasoning:* When a workstation tries to reach a destination outside its own local subnet (such as an internet web server), it needs a designated Layer 3 exit gateway to forward its packets onward to external routers.

3. **Deploy and Assign Domain Name System (DNS) Servers:**
    - *Action:* Assign valid DNS server addresses (either local DNS forwarders or public alternatives like Google's ***8.8.8.8*** or Cloudflare's ***1.1.1.1***) to all client machines.
    - *Reasoning:* Humans browse networks via human-readable URLs (e.g., www.mmu.ac.ke). Applications require DNS resolution to convert those strings into binary IP destinations prior to packet construction.
4. **Implement a Dynamic Host Configuration Protocol (DHCP) Server:**
    - *Action:* Activate a DHCP server (configured on a local server or directly on the router switch) to automatically hand out the IP addresses, subnet masks, default gateways, and DNS settings.
    - *Reasoning:* Manual static addressing of 125 devices introduces massive human margins for IP conflicts and administrative overhead. (Note: The range given, ***172.16.1.1/16***, should ideally use a tighter mask like ***/24*** for 125 hosts, but can run smoothly on the broad default ***/16*** layout).

Question 3

(50 Marks)

> *You are the team leader contracted to design and install a dedicated communication link between the capital city and county offices, which are 250 km apart. There is no existing link, thus starting afresh. The client instructs that the project must be within budget, and be able to sustain a real-time AI-enabled business application. Further, the county office will also serve as a disaster recovery site for the primary server which is in the capital city. Answer the following questions, justifying your answer.*

a. What would you recommend for the following?

(8 Marks)

- **i. Transmission Medium — Single-Mode Fiber Optic (SMF) Cable:** SMF is mandatory for a 250 km span. It uses a narrow glass core and long-wave laser light, allowing signals to travel long distances with negligible attenuation and zero electromagnetic interference (EMI). High-speed dense wavelength division multiplexing (DWDM) equipment can be used to span the distance with minimal intermediate amplification.
- **ii. Transmission Mode — Full-Duplex Mode:** Full-duplex allows data to flow upstream and downstream at the exact same time. Real-time AI execution requires simultaneous bi-directional communication channels for low-latency queries and immediate database state synchronization.
- **iii. Bandwidth — 10 Gbps (Gigabit Ethernet Link) Base Provisioning:** This provides the high throughput needed to feed large real-time data payloads to AI models while managing continuous backup streams to the disaster recovery site.
- **iv. Network Device(s) — Enterprise-Grade Core Edge Routers & Dense Wavelength Division Multiplexing (DWDM) Transponders:** Edge routers at each end manage intelligent traffic prioritization (QoS). DWDM transponders multiplex several optical channels into a single optical fiber pair over long distances, ensuring the 250km span performs reliably.

b. For your answers in (a) above, list two disadvantages for each item.

(8 Marks)

Recommended Item	Disadvantage 1	Disadvantage 2
Single-Mode Fiber	High initial capital expenditure (CapEx) for physical trenching, civil engineering permits, and laying cable across 250 km.	Susceptible to accidental physical cuts from construction or natural disasters, requiring specialized fusion splicing to repair.
Full-Duplex Mode	Requires twice the physical or logical transmission channels (separate send/receive paths), increasing port hardware costs.	Increased software/hardware complexity to manage and prevent buffering bottlenecks or internal cross-talk at transceiver boundaries.
10 Gbps Bandwidth	Higher recurring operational cost (OpEx) for ISP SLA maintenance contracts and high-spec interface transceivers.	Demands high-performance server processing units (NICs, CPUs) to handle data line rates without dropping frames.

Recommended Item	Disadvantage 1	Disadvantage 2
Routers & DWDM Equipment	Complex configuration, troubleshooting, and tuning parameters requiring certified, highly-paid network engineers.	Single point of failure vectors if high-availability redundant hardware pairs are omitted from the design.

c. Which communication protocols will be deployed?

(4 Marks)

The system requires a robust, layered protocol stack to ensure reliable real-time operation and disaster recovery synchronization:

- **BGP (Border Gateway Protocol) / OSPF (Open Shortest Path First):** OSPF handles efficient routing within each office, while BGP manages stable routing and failover across the dedicated 250 km link between the two locations.
- **TCP (Transmission Control Protocol):** Used for database replication and disaster recovery backups. It guarantees in-order, lossless packet delivery through retransmissions and windowing.
- **UDP (User Datagram Protocol) with RTP (Real-time Transport Protocol):** Used for latency-sensitive, real-time AI processing streams and telemetry. This prioritizes speed by stripping away structural connection setup overhead.
- **HTTPS/TLS 1.3 & IPsec:** Encrypts AI API interactions and administrative traffic over the wire to protect sensitive data.

d. Design the addressing that will ensure safe delivery of data through the link. *(10 Marks)*

To establish safe routing boundaries, we will use a structured Variable Length Subnet Masking (VLSM) schema based on the private ***10.0.0.0/8*** class A allocation block. This ensures clean isolation, clear routing paths, and room for future growth:

Network Sector	Subnet Block Assigned	Subnet Mask	Usable IP Range	Use Case Allocation
Capital City Primary HQ	*10.1.0.0/16*	*255.255.0.0*	*10.1.0.1* to *10.1.255.254*	Hosts primary processing nodes, database clusters, and local corporate users.
County Office Disaster Site	*10.2.0.0/16*	*255.255.0.0*	*10.2.0.1* to *10.2.255.254*	Houses hot-standby target recovery targets and regional staff.
Dedicated Inter-Site WAN Link	*10.254.1.0/30*	*255.255.255.252*	*10.254.1.1* (HQ Router) *10.254.1.2* (County Router)	Point-to-point interconnect. Eliminates address waste and prevents routing loops.

Design Implementation Safety Controls:

- **Prevention of Overlapping Address Space:** HQ (***10.1.0.0/16***) and County (***10.2.0.0/16***) are completely distinct subnets. This allows bidirectional communication and real-time replication to work flawlessly without NAT conflicts or routing loops.
- **Strict Edge Summarization:** Core routers will advertise simple summarized routes (***10.1.0.0/16*** and ***10.2.0.0/16***) across the WAN link. This minimizes router memory overhead and accelerates route lookup times.

e. Which measures will be deployed to ensure secure data transmission? *(10 Marks)*

To safeguard data traversing a 250 km physical link, a multi-layered security strategy must be deployed:

1. **IPsec VPN Tunneling (AES-256 GCM Encryption):** Secure all traffic passing between the sites inside an encrypted IPsec tunnel at the router level. This uses AES-256 bit encryption along with SHA-512 hashing to guarantee data confidentiality and integrity. This prevents any unauthorized access or tampering if the fiber line is intercepted.
2. **Deployment of Next-Generation Firewalls (NGFW):** Place firewalls at both ends of the WAN link to inspect incoming and outgoing traffic. They will enforce strict access rules, allowing only valid replication and AI traffic while dropping all other unauthorized connection attempts.
3. **Network Access Control (802.1X):** Use 802.1X authentication to ensure only pre-authorized, trusted corporate servers and devices can connect to the network and transmit data.

4. **Quality of Service (QoS) Prioritization with Bandwidth Reservation:** Configure strict priority queues for real-time AI app traffic to guarantee low latency. At the same time, limit backup traffic to a designated rate to prevent data syncs from overwhelming the link.
5. **Physical Security Infrastructure:** Secure fiber distribution hubs and termination points within locked, biometric-controlled server rooms. Implement continuous optical power monitoring to instantly detect and alert staff to any physical tampering or wiretapping attempts on the fiber optic line.

f. Draw a well labelled diagram depicting your design as answered above. *(10 Marks)*

The network topology diagram below illustrates the structural layout of the high-availability connection linking the Capital City Primary HQ to the County Office Disaster Recovery Site across the 250 km span:

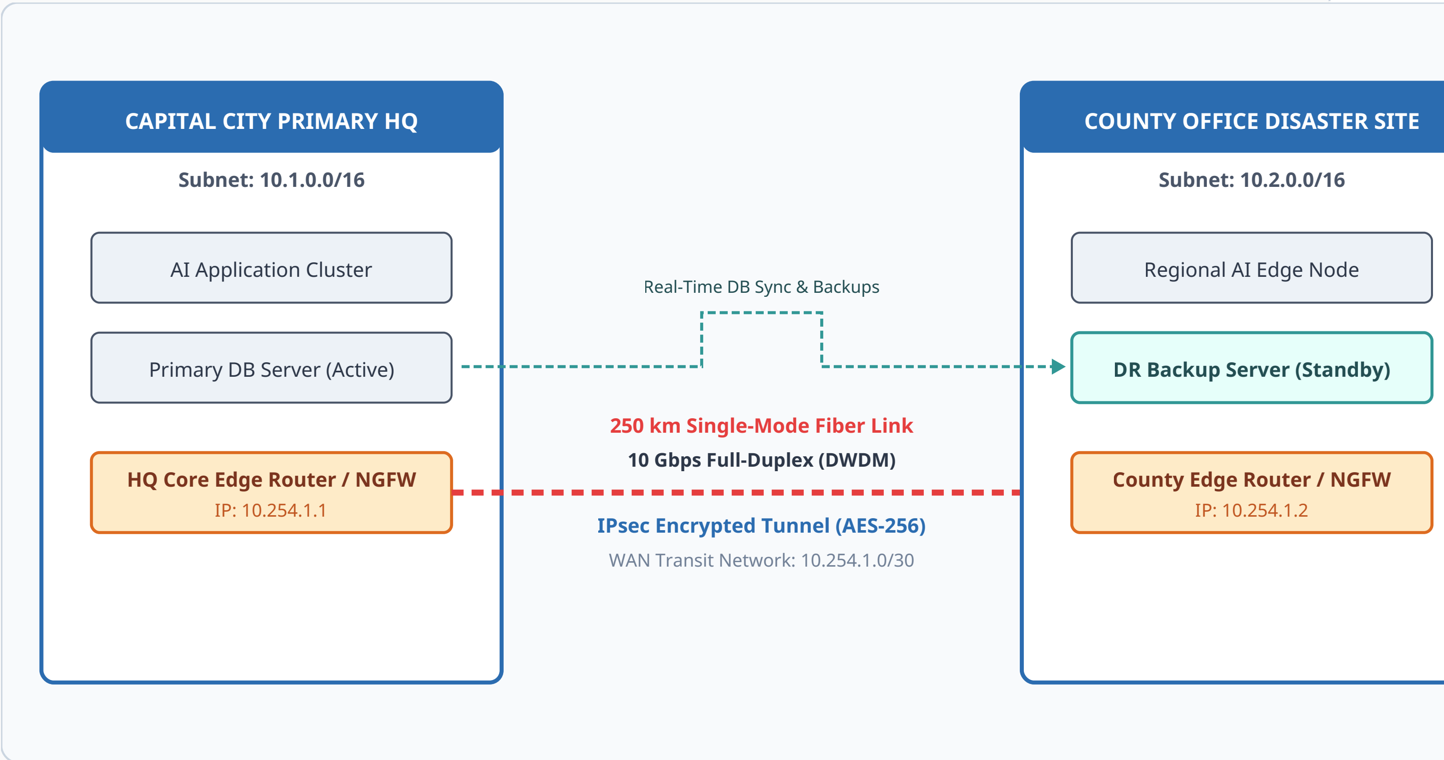

Figure 3.1: Layer 3 Topology Map & Security Boundaries for Capital-County WAN Connection.